

An Equivalent-Key Chosen-Plaintext Break of a 2025 Chaos-Based Image Cipher

Anonymous (single author)
Affiliation withheld for review
author@example.org

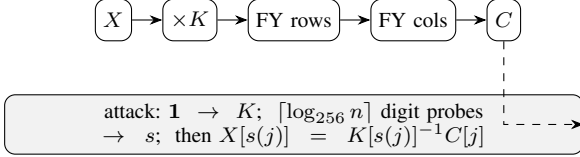


Fig. 1. LLEO encryption (top): key-only multiplicative substitution $\times K$, then two Fisher–Yates block shuffles. The equivalent-key attack (bottom) recovers the multiplier K and source map s from a few chosen plaintexts and inverts the cipher without the key.

Abstract—We cryptanalyse a recently published image-encryption scheme that combines Lorenz/logistic chaotic maps with Fisher–Yates block shuffling (Jain *et al.*, *Optik*, 2025). We observe that the scheme’s keystream and both permutations depend only on the secret key and never on the image, so the whole encryption is a fixed, plaintext-independent map. We reconstruct the cipher from its description—matching its reported near-ideal entropy (e.g. 7.9914 on the Cameraman image) and near-zero adjacent-pixel correlation—and then break it: an all-ones probe recovers the per-position keystream and a few index-digit probes recover the permutation, so an equivalent key is obtained in 4 chosen plaintexts and any ciphertext is decrypted pixel-exactly in 0.027 s without the key. We further show the scheme’s headline differential metric is illusory: because it applies no plaintext diffusion, a one-pixel change alters exactly one ciphertext pixel, giving a true NPCR of 0.000381 % (i.e. $100/MN$), not the ≈ 99.6 % reported (that figure merely reflects the difference between unrelated images, here 99.8833 %). We conclude with the standard fixes.

Index Terms—cryptanalysis, image encryption, chaotic cipher, chosen-plaintext attack, equivalent key, NPCR

I. INTRODUCTION

Chaos-based image ciphers appear in the literature at a high rate, and a recurring lesson from the cryptanalysis side is that a large fraction of them succumb to the same structural attack. Most follow a permutation–diffusion template: a confusion stage that rearranges pixels and a diffusion stage that masks pixel values with a chaotic keystream. When that keystream is a function of the secret key alone and not of the image, the entire cipher collapses to a *fixed* permutation–substitution map, and an *equivalent key*—a representation that decrypts every ciphertext under the same secret key—can be recovered from a handful of chosen plaintexts [1]–[4]. The defence is equally well known: the keystream must depend on the plaintext, and the cipher must actually diffuse [5].

A second, persistent problem is methodological. Authors validate schemes with statistical scores—Shannon entropy, adjacent-pixel correlation, and the differential metrics NPCR and UACI—and treat near-ideal values as evidence of security. None of these scores measures resistance to a chosen-plaintext attack, and, as we show, all of them can be near-ideal for a cipher that is broken in seconds.

This paper applies both lessons to a 2025 scheme [6] (which we abbreviate LLEO) built from the Lorenz [7] and logistic maps with Fisher–Yates block shuffling (Fig. 1). Our contributions are:

- We identify that LLEO is entirely key-driven and therefore a fixed monomial map over $\mathbb{Z}_{256}$ (Section III–IV).
- We give an equivalent-key chosen-plaintext attack that recovers the full map in $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts—4 for a 512×512 image—and decrypts any image pixel-exactly without the key (Section IV).
- We reproduce LLEO, confirm it matches the paper’s reported statistics, and then show those statistics coexist with *zero* plaintext diffusion: the true one-pixel NPCR is 0.000381 %, not 99.6 % (Section V).

Fig. 2 previews the outcome: four standard images, their LLEO ciphertexts, and our key-free recoveries, each bit-identical to the original and obtained in 4 chosen plaintexts.

II. RELATED WORK

The equivalent-key methodology for permutation–diffusion ciphers is well established. Arroyo *et al.* [3] broke a “total shuffling” scheme by recovering its key-only keystream; Li *et al.* [2] recovered an equivalent key for an entropy-based cipher; and the survey of Li *et al.* [1] catalogues a year of such breaks, nearly all exploiting plaintext-independent diffusion. For the confusion stage alone, Li and Lo [4] established the optimal number of known/chosen plaintexts needed to invert a permutation-only multimedia cipher, of order $\log_L(MN)$ for L intensity levels—the same logarithmic regime our attack achieves; the general quantitative theory of permutation-only cryptanalysis [8] bounds this effort, and recent chosen-plaintext breaks continue to recover equivalent keys from a handful of images [9], [10], confirming the vulnerability remains a live concern. Alvarez and Li [5] set out basic requirements for chaos-based cryptosystems, including plaintext-dependent keystreams and sensitivity to the plaintext; LLEO violates these. The differential metrics we scrutinise were standardised by Wu *et al.* [11], who also note

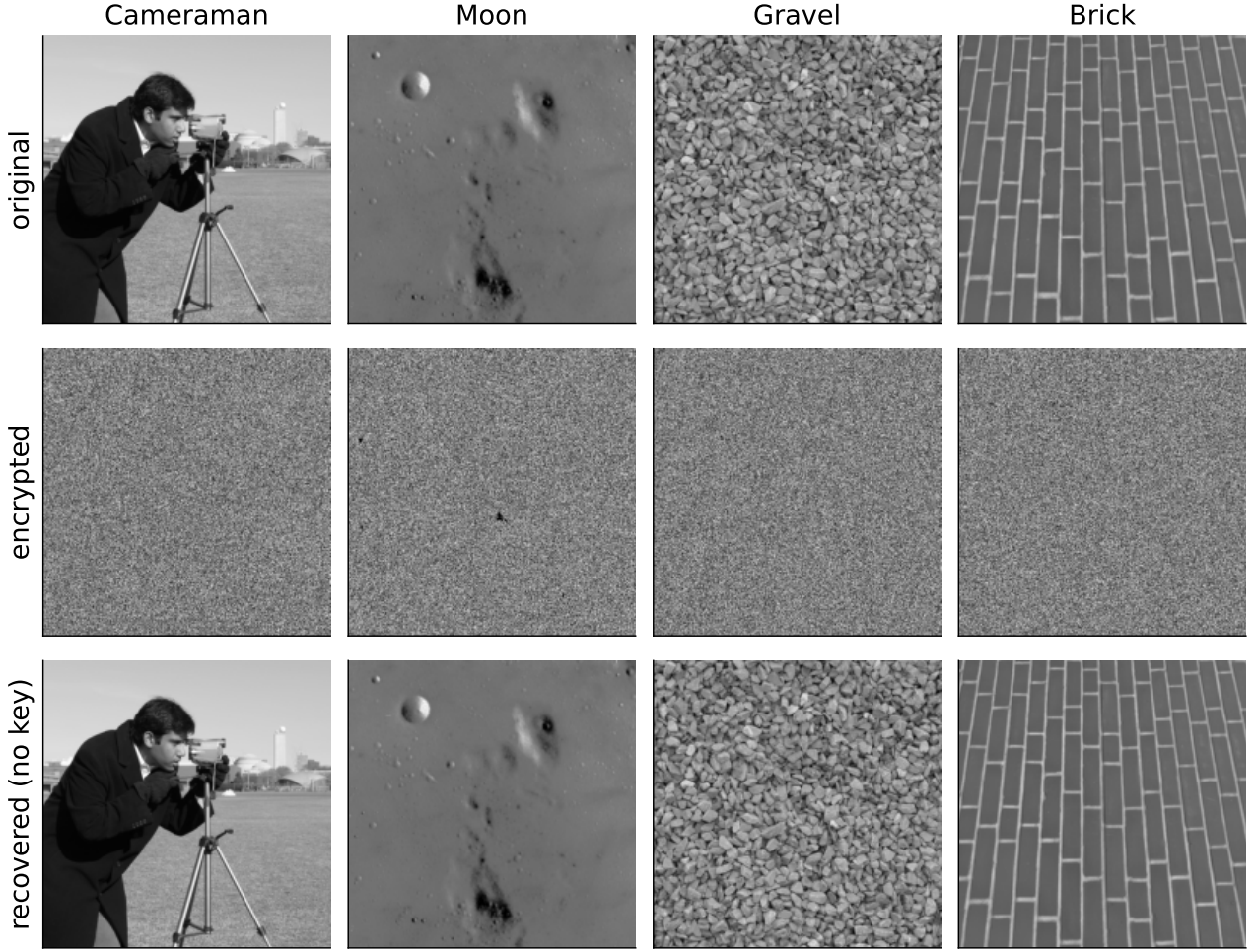


Fig. 2. LLEO original, ciphertext, and key-free recovery for four standard 512×512 images. Each recovery uses only 4 chosen plaintexts and no key, and is bit-identical to the original.

that NPCR/UACI quantify sensitivity to a one-pixel plaintext change—precisely the quantity LLEO fails to deliver despite its reported value.

The pattern is structural, not incidental: when both confusion and diffusion are fixed by the key the cipher is—as a function of the plaintext—an affine or monomial map, and composing more such stages changes only its coefficients [1], [4]. The decisive question is binary: does any key material depend on the image? Schemes that answer “no” (LLEO included) fall to an equivalent key, while those that bind the keystream to a per-image value resist—both sides of which our positive control demonstrates on the same construction (Section V).

III. THE TARGET CIPHER

LLEO encrypts an $M \times N$ grayscale image X in three key-driven steps.

(1) *Substitution*. X is split into 16 horizontal strips. The eight odd strips are combined with a keystream $k_{\log}$ produced by the logistic map $x_{n+1} = r x_n(1 - x_n)$, and the eight even strips with a keystream k_{lor} produced by the Lorenz system

$\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$, through an invertible per-position operation; decryption uses the modular “conjugate” of the keystream, i.e. a multiplicative inverse.

(2) *Permutation 1*. The 16 horizontal strips are reordered by a Fisher–Yates shuffle.

(3) *Permutation 2*. The result is divided into 16 vertical strips and shuffled again, giving the ciphertext.

The four “decryption keys” are the two keystreams (d_1, d_2) and the two shuffle index lists (d_3, d_4) . The chaotic initial conditions are derived from the secret key and the shuffles are seeded from it; *no quantity depends on the image X* . Although the scheme is repeatedly called “asymmetric,” the same four keys both encrypt and decrypt: it is a symmetric cipher.

IV. CRYPTANALYSIS

A. Threat model

We work in the standard chosen-plaintext setting under Kerckhoffs’s principle: the attacker knows the algorithm, has oracle access to encryption under a fixed unknown key, and seeks an *equivalent key*—a procedure that decrypts any ciphertext of the probed size—not the chaotic secret seed

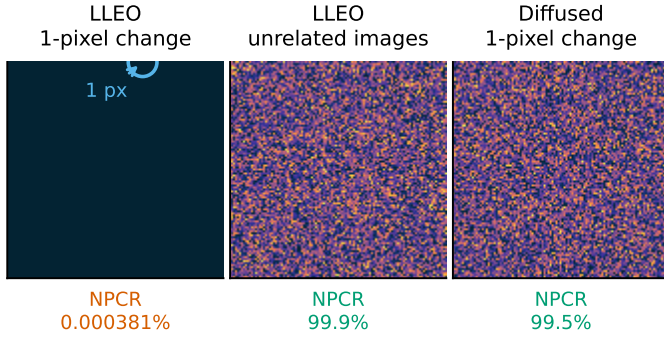


Fig. 3. The no-diffusion property, made visible (used quantitatively in Section V): the absolute ciphertext difference $|C_0 - C_1|$ (dark = unchanged, bright = changed), under one fixed key. **(left)** A one-pixel plaintext change alters *exactly one* ciphertext pixel (ringed)—no diffusion—so the true one-pixel NPCR is $100/MN \approx 0.000381\%$, not the reported $\approx 99.6\%$. **(centre)** That $\approx 99.6\%$ is really the difference between two *unrelated* images (here 99.8833%), which any near-uniform cipher attains. **(right)** Seeding the keystream from SHA-256 of the plaintext diffuses the *same* one-pixel change into full noise—what a one-pixel NPCR should measure.

itself. Because the cipher is plaintext-independent, the same equivalent key also follows from enough known plaintext–ciphertext pairs [12]; we present the chosen-plaintext version as it is minimal.

B. Structural observation

Index the image as a length- n vector, $n = MN$. Each LLEO step is linear over $\mathbb{Z}_{256}$ and fixed by the key: substitution multiplies coordinate p by a fixed unit $K[p]$ (odd, hence invertible mod 256), and the two block shuffles compose into one position permutation P . Therefore

$$C[j] = K[s(j)] \cdot X[s(j)] \pmod{256}, \quad s = P^{-1}, \quad (1)$$

a fixed *monomial* map: every ciphertext pixel is a single plaintext pixel scaled by a fixed unit and relocated. Two consequences are immediate. There is no additive constant, since $E(\mathbf{0}) = \mathbf{0}$. And there is no diffusion: $C[j]$ depends on exactly one input coordinate $s(j)$ —so a one-pixel plaintext change alters exactly one ciphertext pixel, which we exploit in Section V and visualize in Fig. 3.

Proposition 1. *If a permutation–diffusion cipher’s keystream and permutations depend only on the key, it realises a fixed affine map of the plaintext: the monomial (1) with each $K[p]$ a unit in $\mathbb{Z}_{256}$ when the substitution is a modular multiply, or an affine map $C = Mx \oplus b$ over $\text{GF}(2)$ (with $b = E(\mathbf{0})$ possibly nonzero) when the substitution is XOR-based; composing any number of such rounds yields a map of the same form. In either case it is recovered from $1 + \lceil \log_{256} n \rceil$ chosen plaintexts: one all-ones (resp. all-zeros) query fixes the substitution at every output position, and $\lceil \log_{256} n \rceil$ index-digit queries fix the permutation s .*

Proof sketch. Each stage—multiplication by the key-derived K and the two block shuffles—is a fixed function of the key applied to pixel values or positions, so any composition is a fixed key-only map; over $\mathbb{Z}_{256}$ this is the monomial (1).

Algorithm 1 Equivalent-key recovery for LLEO

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1:  $\hat{K} \leftarrow \text{Enc}(\mathbf{1}) \triangleright \hat{K}[j] = K[s(j)]$ : multiplier at output  $j$ 
2: for  $t \leftarrow 0$  to  $\lceil \log_{256} n \rceil - 1$  do
3:    $C \leftarrow \text{Enc}(P_t) \triangleright P_t[i] =$  the  $t$ -th base-256 digit of index  $i$ 
4:    $d_t[j] \leftarrow C[j] \cdot \hat{K}[j]^{-1} \pmod{256}$  for all  $j$ 
5: end for
6:  $s[j] \leftarrow \sum_t d_t[j] 256^t \triangleright$  source index for each output  $j$ 
7: return  $\text{Dec}(C) : X[s(j)] \leftarrow \hat{K}[j]^{-1} C[j] \pmod{256}$ 

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The keystream is forced odd, so $K[p]$ is a unit and $x \mapsto K[p]x$ is a bijection with inverse $K[p]^{-1}$. Querying $X = \mathbf{1}$ gives $C[j] = K[s(j)]$; querying the t -th base-256 digit of the index gives $C[j] = K[s(j)] \text{digit}_t(s(j))$, and dividing by the recovered $K[s(j)]$ yields $\text{digit}_t(s(j))$; collecting all digits reconstructs s . If instead the substitution is XOR with a key-only keystream, the same argument over $\text{GF}(2)$ gives an affine map $C = Mx \oplus b$ with $b = E(\mathbf{0})$: the all-zeros query returns b and the index-digit queries return s , exactly as in the CBC-like XOR cipher we break in Section IV. $\square$

C. Equivalent-key recovery

Because K and s are fixed, two kinds of chosen plaintext suffice.

(a) *Multiplier, one query.* Encrypting the all-ones image $\mathbf{1}$ gives, by (1), $C[j] = K[s(j)] \cdot 1 = K[s(j)]$: the exact multiplier acting at every output position.

(b) *Permutation, $\lceil \log_{256} n \rceil$ queries.* Let P_t be the plaintext whose pixel i holds the t -th base-256 digit of its index i . Then $C[j] = K[s(j)] \cdot \text{digit}_t(s(j))$; multiplying by the already-known $K[s(j)]^{-1}$ yields $\text{digit}_t(s(j))$. Collecting all $\lceil \log_{256} n \rceil$ digits reconstructs $s(j)$ for every j , hence the full permutation.

The recovered pair (s, K) is an equivalent key; any ciphertext is inverted by $X[s(j)] = K[s(j)]^{-1} C[j]$. The total cost is

$$1 + \lceil \log_{256} n \rceil \text{ chosen plaintexts}, \quad (2)$$

i.e. 4 for a 512×512 image, independent of the nominal key length.

Concretely, a 512×512 image needs four queries—one all-ones image and three index-digit images—after which Algorithm 1 returns the full map in a single pass. Table I states the complexity and compares it to the optimal permutation-only bound of Li and Lo [4]: our query count is order-optimal and the cost is independent of the cipher’s nominal key length, so the advertised key space is irrelevant—the equivalent key bypasses key recovery entirely [13].

D. Why extra structure does not help

A composition of fixed key-only maps is again one, so extra rounds, chaotic maps, or block shuffles leave (1) intact (only K, s change) and the attack applies unchanged; it is moreover a *known-plaintext* attack whenever the available plaintexts span the index digits. The single property that would defeat it is a keystream that depends on X .

TABLE I
ATTACK COMPLEXITY VS. THE OPTIMAL PERMUTATION-ONLY BOUND
($n = MN$ PIXELS, $L = 256$ LEVELS).

	This attack	Optimal perm.-only [4]
Data (chosen plaintexts)	$1 + \lceil \log_{256} n \rceil$	$\lceil \log_L n \rceil$
Time	$O(n \lceil \log_{256} n \rceil)$	$O(n \lceil \log_L n \rceil)$
Memory	$O(n)$	$O(n)$

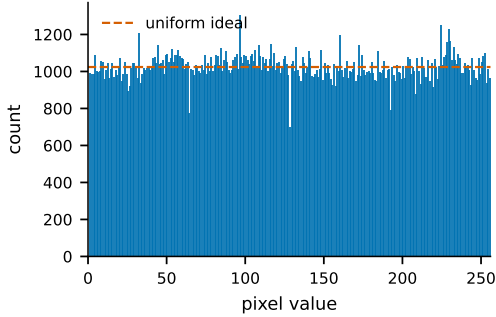


Fig. 4. Cipher histogram (Cameraman): near-uniform, yet the cipher is trivially broken.

V. EXPERIMENTAL RESULTS

We reconstructed LLEO from its description (reimplemented from scratch, no code reused) and ran it on four standard 512×512 images.

A. Reproduction

Table II shows the reconstruction reproduces LLEO’s claimed statistics: cipher entropy approaches the ideal 8 (Cameraman 7.9914, up from the plaintext’s 7.2317) and adjacent-pixel correlation drops to near zero (-0.0073). Fig. 4 shows the flat cipher histogram and Fig. 5 the destroyed correlation. By the usual yardsticks the cipher looks strong.

B. The break

The attack of Section IV recovers the equivalent key in 4 chosen plaintexts and decrypts a fresh ciphertext pixel-exactly in 0.027 s (Cameraman), using no key.¹ Fig. 2 shows the original, the LLEO ciphertext, and the key-free recovery, which is identical to the original.

C. Cost scales logarithmically

Because the attack needs only $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts, its cost is nearly flat as the image grows (Table III): 3 plaintexts at 128^2 , 4 at 512^2 , and still 4 at 1024^2 , each recovered exactly in well under a second. The key length the scheme advertises does not enter this cost.

D. The differential metric is illusory

LLEO reports $\text{NPCR} \approx 99.6\%$ as evidence of differential-attack resistance. But by (1) flipping one plaintext pixel changes exactly one ciphertext pixel (Fig. 3, left), so the

¹Wall-clock, single core. The 512^2 entry in Table III (0.037 s, a random image) differs only by run-to-run variation.

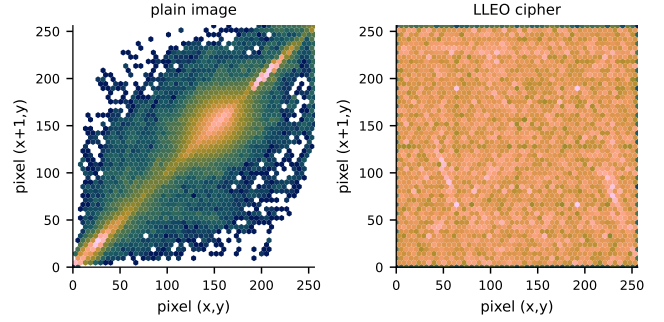


Fig. 5. Adjacent-pixel correlation as a hexbin density (every horizontal pixel pair, log counts): the plain image concentrates on the diagonal—neighbours are alike—while the LLEO cipher fills the plane uniformly (Cameraman).

TABLE II
RECONSTRUCTION STATISTICS AND ATTACK OUTCOME (512×512).
EVERY IMAGE IS RECOVERED PIXEL-EXACTLY WITHOUT THE KEY.

Image	H plain	H cipher	corr. cipher	chosen PT	broken
Cameraman	7.2317	7.9914	-0.0073	4	yes
Moon	4.885	7.9316	0.0011	4	yes
Gravel	7.2531	7.9876	-0.0005	4	yes
Brick	5.4553	7.8966	-0.0007	4	yes

true one-pixel NPCR is $100/MN = 0.000381\%$ (UACI 4e-06%)—five orders of magnitude below the reported value. The 99.6% figure instead measures the difference between *unrelated* images, which for LLEO is 99.8833% (Fig. 3, centre); this is achieved by any cipher with a near-uniform output and says nothing about chosen-plaintext security. Table IV contrasts each claimed property with what the cipher actually provides. Finally, the chaotic keystreams are themselves non-uniform (Fig. 6): a chi-square test rejects uniformity for the logistic keystream ($\chi^2 = 525839.6$, $p < 10^{-300}$, numerically zero).

E. The attack is not specific to LLEO

The same principle breaks any key-only permutation-diffusion cipher. As a second, structurally distinct example we take the canonical construction with a CBC-like XOR diffusion chain, $C[i] = X_\pi[i] \oplus k[i] \oplus C[i-1]$ —representative of a large published family and unlike LLEO’s multiplicative monomial form. With a key-only keystream this map is affine over $\text{GF}(2)$ (the identity $E(a) \oplus E(b) \oplus E(0) = E(a \oplus b)$ holds at one *and* three rounds in our checks), and the equivalent-key CPA recovers it exactly in 3 chosen plaintexts (one all-zeros query plus $\lceil \log_{256} n \rceil$ digit probes—the same $1 + \lceil \log_{256} n \rceil$ rule). The identical procedure fails against a plaintext-hash-seeded version of the *same* structure, reinforcing that what matters is the keystream’s dependence on the image—not the algebra or the round count.

VI. DISCUSSION

LLEO illustrates two persistent failure modes. First, *good statistics are not security*: a flat histogram, near-ideal entropy, near-zero correlation, and a large inter-image NPCR are all

TABLE III
ATTACK COST VS. IMAGE SIZE (RANDOM PLAINTEXT, SINGLE CORE).

Size	chosen PT	time (s)	recovered
128×128	3	0.003	exact
256×256	3	0.009	exact
512×512	4	0.037	exact
1024×1024	4	0.254	exact

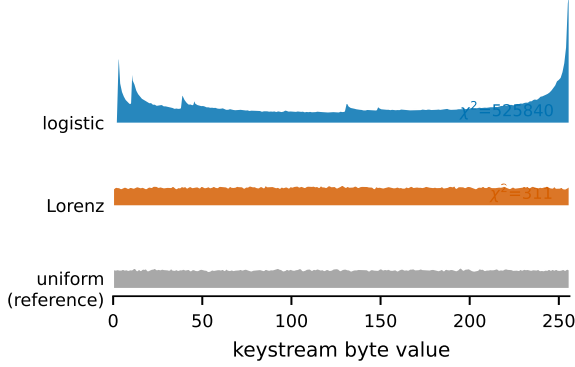


Fig. 6. Keystream byte distributions (ridgeline). The logistic keystream is sharply non-uniform—its invariant density spikes at the extremes—and even the Lorenz stream is measurably lumpy, whereas a secure stream cipher would match the flat uniform reference. A chi-square test rejects uniformity for both (χ^2 annotated).

satisfied by our fixed, key-only map, which we nonetheless break in 4 queries—testing plaintext sensitivity directly [5], [11] is what catches this. Second, *naming is not architecture*: labelling a symmetric construction “asymmetric” adds no security.

The remedies are standard. (i) Make the keystream depend on the plaintext—e.g. derive the chaotic initial conditions from a cryptographic hash (SHA-256) of the image, so that a different plaintext yields an unrelated keystream and the all-ones/digit probes no longer reveal a reusable equivalent key. (ii) Add genuine diffusion, e.g. a chaining rule $C[i] = f(X[i], k[i], C[i-1])$, so that one plaintext change propagates and the true NPCR approaches its ideal. With either defence in place the map of (1) is no longer fixed across images and the attack of Section IV fails. We verified this directly: the identical attack recovers 100.0 % of pixels against the key-only cipher but only 0.2518 % (chance level) once the keystream is seeded from a SHA-256 hash of the image.

Scope and disclosure. We recover an equivalent key that decrypts any ciphertext of the probed size, not the underlying chaotic secret seed—the standard, defensible claim in this literature [1]. Consistent with responsible-disclosure practice, the authors of the target scheme will be notified before camera-ready, and our reproduction and attack code is released to support verification.

VII. CONCLUSION

A 2025 chaos-based image cipher is completely broken by an equivalent-key chosen-plaintext attack in 4 chosen

TABLE IV
CLAIMED VS. ACTUAL SECURITY OF LLEO.

Property	Reported	Actual
Entropy (Cameraman)	≈ 7.997	7.9914 (genuine)
NPCR (1-pixel change)	$\approx 99.6\%$	0.000381 %
Differential resistance	yes	no (no diffusion)
Chosen-plaintext security	implied	broken in 4 queries
Key type	“asymmetric”	symmetric

plaintexts and a fraction of a second, despite near-ideal entropy and a reported NPCR of $\approx 99.6\%$. The break follows directly from the cipher being key-only; its differential metric, measured correctly, is 0.000381 %. We recommend plaintext-dependent keystreams and real diffusion, and that statistical scores accompany—rather than replace—an explicit chosen-plaintext security argument.

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